

MINI-ASSESSMENT

ADVANCED MULTI-STEP PROMPT WORKFLOW

AI Personalized Learning & Study Coach

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1. Objective

The objective of this workflow is to design a multi-step Generative AI prompt workflow that converts a learner's topic, current knowledge level, learning goals, and available study time into a personalized learning plan.

The workflow uses **prompt chaining, task decomposition, personalization, structured generation, adaptive evaluation, and iterative refinement.**

Instead of generating a generic study plan with a single prompt, the system analyzes the learner, decomposes the subject, creates a customized plan, generates learning material, evaluates understanding, and adapts the plan based on performance.

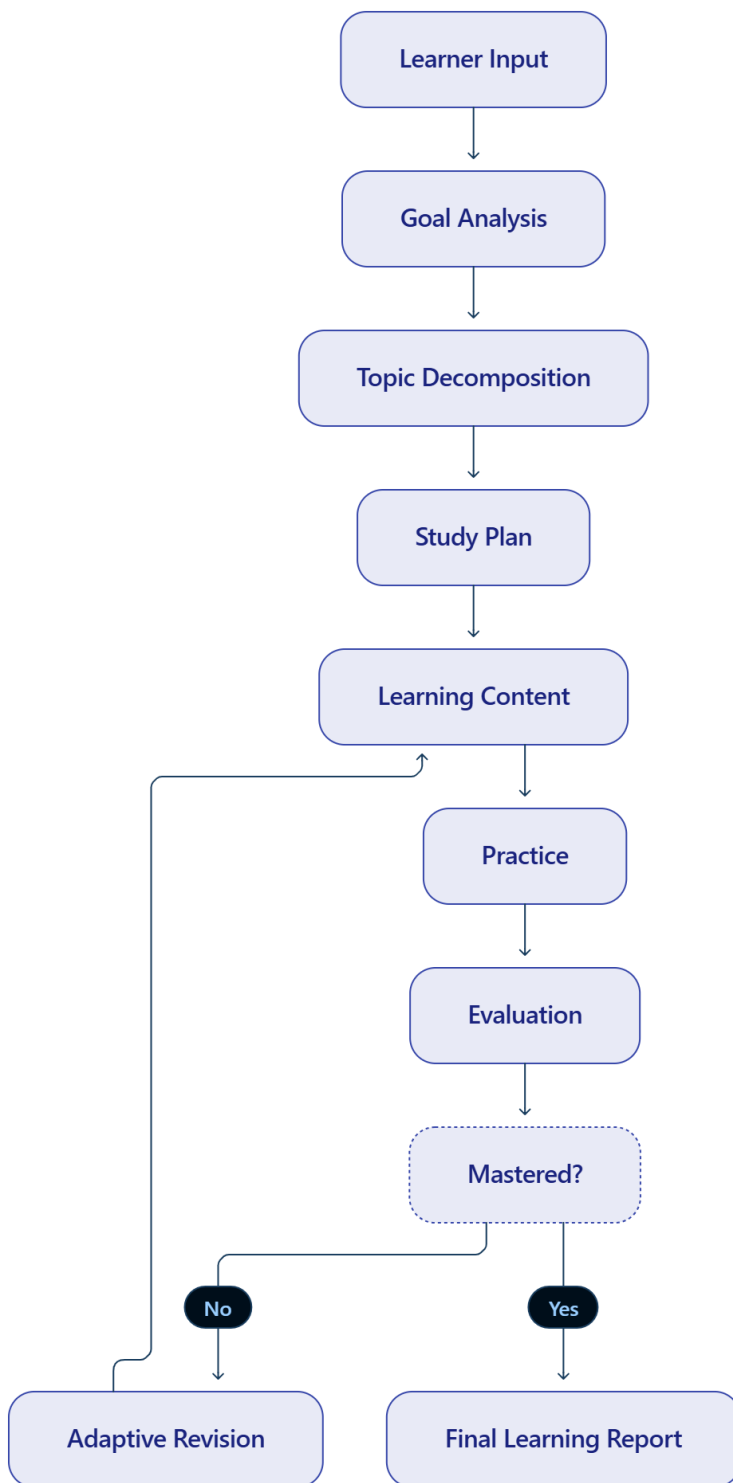
2. Problem Statement

Students often struggle to create effective study plans because they may not know:

- What topics to study first
- Which concepts are prerequisites
- How much time to allocate
- Which areas need more practice
- Whether they actually understand the material

A Generative AI learning coach can address these problems through a multi-step workflow that continuously adapts to the learner.

3. Workflow Overview



4. Step 1 – Learner and Goal Analysis

Purpose

Understand the learner's current knowledge, objective, constraints, and preferred learning approach.

Prompt

You are an AI Learning Strategy Assistant.

Analyze the learner's input and identify:

1. Learning goal
2. Current knowledge level
3. Target knowledge level
4. Available study time
5. Learning deadline
6. Topics the learner already understands
7. Topics the learner finds difficult
8. Preferred learning approach
9. Expected outcome

Identify missing information or assumptions.

Do not generate the complete study plan yet.

Return a structured learner profile.

Output

Personalized Learner Profile

Step 2 – Topic Decomposition

Purpose

Break the learning objective into concepts and identify prerequisite relationships.

Prompt

Using the learner profile from Step 1, decompose the target subject into a structured learning hierarchy.

Identify:

1. Core concepts
2. Subtopics
3. Prerequisite concepts
4. Beginner-level concepts
5. Intermediate concepts

6. Advanced concepts
7. Topics requiring practical exercises

Organize the topics in a logical learning order.

Do not create the final study schedule yet.

Output

Learning Topic Map

Step 3 – Generate Personalized Study Plan

Purpose

Create a study schedule based on the learner's available time and target.

Prompt

Using the learner profile and topic map, create a personalized study plan.

Allocate study time according to:

- Topic difficulty
- Prerequisites
- Learner's current knowledge
- Importance to the learning goal
- Available study time
- Learning deadline

For each study session provide:

Topic | Objective | Duration | Activity | Expected Outcome

Include revision and practice sessions.

Prioritize weak areas while maintaining prerequisite order.

Output

Personalized Study Schedule

Step 4 – Generate Learning Content

Purpose

Generate learning material appropriate to the learner's current level.

Prompt

Using the personalized study plan, generate learning material for the selected topic.

Adapt the explanation to the learner's current knowledge level.

Include:

1. Simple explanation
2. Key concepts
3. Practical examples
4. Common mistakes
5. Important terminology
6. Short summary
7. One real-world application

Do not introduce advanced concepts before explaining their prerequisites.

Output

Personalized Learning Module

Step 5 – Generate Practice and Quiz

Purpose

Test whether the learner understands the material.

Prompt

Using the learning module from Step 4, create a practice assessment.

Generate:

- 3 easy questions
- 3 medium questions
- 2 challenging questions
- 2 practical/application questions

Include an answer key separately.

Questions should test understanding and application rather than simple memorization.

Do not reveal answers alongside the questions.

Output

Adaptive Practice Assessment

Step 6 – Performance Evaluation

Purpose

Evaluate the learner's performance and identify weak areas.

Prompt

Evaluate the learner's answers against the assessment from Step 5.

Calculate:

1. Overall score
2. Concept-level performance
3. Strong areas
4. Weak areas
5. Common mistakes
6. Concepts requiring revision

Classify each concept as:

- Mastered
- Developing
- Needs Revision

Provide evidence for each classification.

Do not assume understanding when the learner's answer does not demonstrate it.

Output

Learning Performance Report

10. Adaptive Decision and Feedback Loop

Decision

The workflow evaluates whether the learner has achieved the required learning target.

If the learner has NOT achieved the target:

Performance Evaluation → Identify Weak Areas → Adaptive Revision → New Learning Content → Practice → Evaluation

If the learner HAS achieved the target:

Performance Evaluation → Final Learning Summary

This creates an **adaptive learning loop** rather than a fixed study plan.

Step 7 – Adaptive Revision

Purpose

Regenerate learning material specifically for concepts the learner has not mastered.

Prompt

Using the learner's performance report, identify the concepts classified as "Needs Revision" or "Developing."

Create a targeted revision module for those concepts.

For each weak concept:

1. Explain the concept differently from the previous explanation.
2. Provide an easier example.
3. Identify the learner's likely misunderstanding.
4. Provide a practical exercise.
5. Generate one checkpoint question.

Do not repeat concepts that have already been classified as mastered unless they are prerequisites for a weak concept.

Output

Targeted Revision Module

Step 8 – Final Learning Summary

Purpose

Create a final overview of the learner's progress.

Prompt

Using the learner profile, study plan, learning modules, practice results, and performance evaluations, create a final learning progress report.

Include:

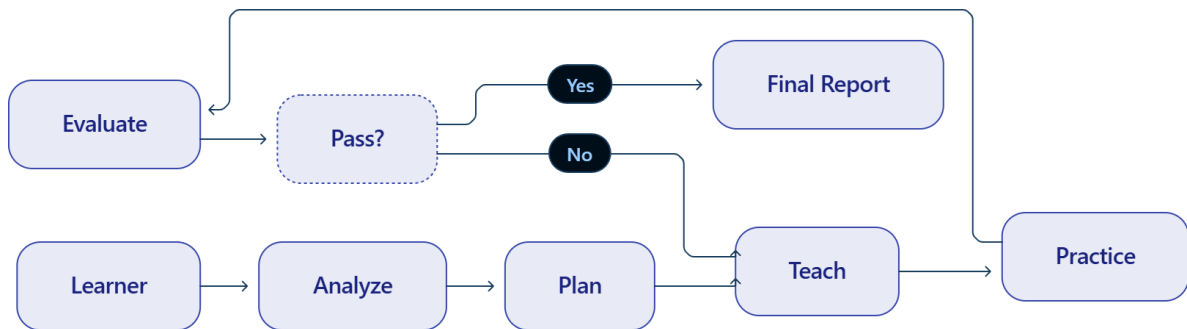
1. Original learning goal
2. Topics mastered
3. Topics still developing
4. Topics requiring further revision
5. Overall performance
6. Recommended next steps
7. Suggested future learning path

Clearly distinguish demonstrated knowledge from recommended future study.

Output

Final Personalized Learning Report

6. Detailed AI Workflow



7. Input and Output Flow

Step	Input	Processing	Output
1	Learner information	Goal analysis	Learner profile
2	Learner profile	Topic decomposition	Topic map
3	Topic map + constraints	Plan generation	Study schedule
4	Study plan	Content generation	Learning module
5	Learning module	Question generation	Practice assessment
6	Learner answers	Performance analysis	Performance report
7	Weak areas	Adaptive generation	Revision module
8	Learning history	Progress evaluation	Final learning report

8. Example Input

Learner Request

I want to learn Machine Learning for an internship interview. I know Python and basic statistics. I have 10 days and can study 2 hours per day. I find algorithms and model evaluation difficult.

Expected AI Behavior

The workflow should:

1. Identify Python and basic statistics as existing knowledge.
2. Identify machine learning fundamentals as the target.
3. Prioritize algorithms and model evaluation because they are weak areas.
4. Create a 10-day study plan.
5. Allocate approximately 2 hours per day.
6. Generate explanations and practical examples.
7. Test understanding with progressively difficult questions.
8. Identify weak concepts from the answers.
9. Generate targeted revision.
10. Produce a final progress report.

9. Key Generative AI Concepts Demonstrated

1 Prompt Chaining

Each stage uses outputs generated by previous stages.

2 Task Decomposition

The complex learning problem is divided into analysis, planning, generation, testing, and evaluation.

3 Personalization

The AI adapts content based on the learner's current knowledge, goal, difficulty areas, and time constraints.

4 Structured Prompting

Each prompt defines a role, objective, constraints, and expected output.

5 Adaptive Generation

Learning content changes according to the learner's performance.

6 Classification

Concepts are classified as **Mastered, Developing, or Needs Revision**.

7 Iterative Refinement

Weak areas trigger targeted revision and another evaluation cycle.

8 Evaluation

Learner performance is measured using defined criteria.

9 Feedback Loop

The workflow uses evaluation results to influence the next generation stage.

10. Advantages of the Workflow

1. Provides personalized learning instead of generic content.
2. Saves time when creating study plans.
3. Adapts to different knowledge levels.
4. Identifies weak concepts systematically.
5. Provides targeted revision.
6. Uses practice-based evaluation.
7. Supports continuous improvement.
8. Maintains context between workflow stages.
9. Can be reused for different subjects.
10. Demonstrates an adaptive GenAI workflow rather than simple text generation.

11. Expected Final Result

The workflow produces a personalized learning report containing:

- Learner profile
- Learning objective
- Topic dependency map
- Personalized study schedule
- Learning materials
- Practice questions
- Performance analysis
- Weak-area identification
- Adaptive revision
- Final progress assessment
- Recommended next learning path

12. Why This Is a Multi-Step GenAI Workflow

The workflow is not a collection of independent prompts.

Each stage depends on the output of previous stages:

Learner Profile → Topic Map → Study Plan → Learning Content → Practice → Evaluation → Adaptive Revision → Re-evaluation → Final Report

The feedback loop makes the workflow dynamic and allows the AI to modify its behavior based on new information.

13. Conclusion

The AI Personalized Learning and Study Coach demonstrates how Generative AI can be designed as an adaptive multi-step workflow.

The system first understands the learner, decomposes the target subject, creates a personalized study plan, generates appropriate learning material, tests the learner's understanding, evaluates performance, and adapts the learning content based on identified weaknesses.

The feedback loop allows the workflow to continuously refine the learning experience until the desired learning target is achieved.

This workflow demonstrates key GenAI concepts including **prompt chaining, task decomposition, personalization, structured prompting, information extraction, classification, adaptive generation, evaluation, and iterative refinement.**